

How did VxWorks navigate the critical and delicate trajectory correction manoeuvres?

Application programmers at the Jet Propulsion Laboratory (JPL) created packages that allowed the craft to take measures of its current trajectory and apply corrections via rockets, in real time, all the way down to the ground. These packages included timing for triggering pyrotechnic devices to perform various manoeuvres—like ejecting the heat shield or deploying the space crane—exactly when they were needed.

VxWorks is a deterministic OS, which means it allows programmers to control the computer with split-second accuracy. This kind of accuracy is critical to create systems that react to their environment in real time.

How were the avionics and scientific features taken care of?

This is an excellent question for engineers at NASA/JPL. Just as teams at Wind River are experts at tailoring their OS for their customers' needs, Wind River's customers are experts at creating their devices and applications. My job, as an engineer, is to assist our customers in understanding and leveraging our OS to create their applications in a timely and affordable fashion.

The teams at JPL have been creating robotic spacecraft for decades, and are expert at integrating avionics and science packages in the creation of these fantastic explorers.

What challenges did you face when controlling a sequence millions of kilometres into space?

For our OS, the challenges are much the same as with any remote autonomous system—our OS has to perform as promised. Scientists and engineers creating space-bound craft not only have to meet all the hard requirements of earthbound industrial and medical robots but their systems are also going to be unavailable for inspection or debugging once they are deployed. Also, they have to operate in a radiation-filled vacuum under extreme temperature variations after being placed under unimaginable acceleration and vibration!

How do you ensure reliability of your software in such applications?

Testing is a complex subject. Wind River uses a variety of tests, including automated test suites that it has developed and maintained for over two decades. Tests cover various areas including proper functions and error conditions handling.

With automated tests, results are collected and compared against expected results, and reports generated. These tests are separated out into tests specific to hardware interface functions (the BSP part), and general operating system and programming interface tests. ●

(The complete interview can be viewed at electronicsforu.com)

RECOM

Switching Regulators with 40% price reduction



SIP3 & SMD Packages

- Efficiency up to 92%
- 3.3V, 5V / 500mA output
- Ultra wide input voltage range
- Low profile case
- 3-Year warranty

Learn more about RECOM products at www.recomasia.com
Email: enquiry@recomasia.com | Tel: +65 6276-8795

Authorised Local Distributor:



Omniscient
Electronics Pvt. Ltd.,



Team Technology